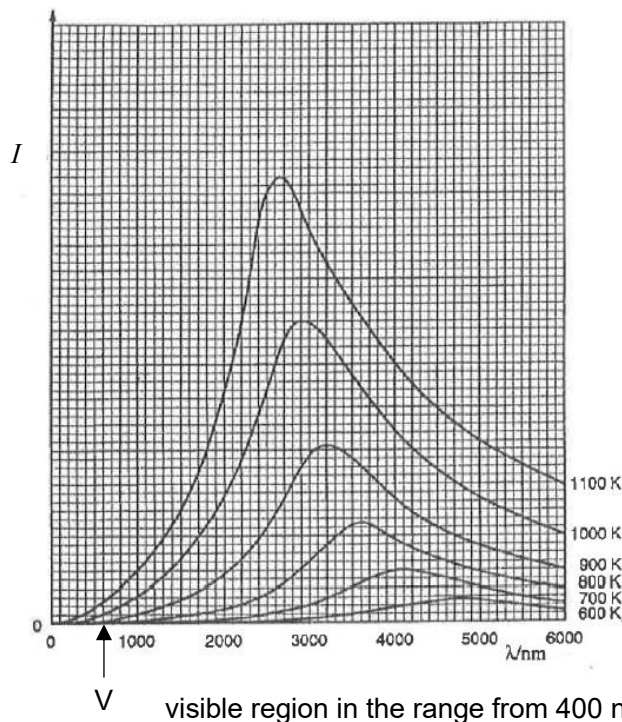


8 (a) Energy has discrete/fixed values.

B1

Comments: Generally poorly answered. Use of the word packet does not connote quantised.

(b) (i)



- (ii) In the visible region, the intensity of the emitted radiation increases with increasing wavelength. **B1**

The red region has the longest wavelength and according to the graph highest intensity at 1100 K **B1**

The hot object is perceived by the eye as glowing red. **A0**

- (c) (i) The wavelength, $\lambda_{\max}$, that corresponds to the peak intensity of the emitted radiation. The higher the temperature, the shorter the $\lambda_{\max}$. **B1**

- (ii) advantage: The radiation can be detected at a distance. Hence there is no need for contact between device and the body to measure its temperature. **B1**

disadvantage: At lower temperature, the peak cannot be easily identified. **B1**

- (d) (i) Electric force of attraction on electron provides the centripetal force for its motion about the nucleus. **B1**

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r} \quad \text{C1}$$

$$v = \sqrt{\frac{e^2}{4\pi\epsilon_0 mr}} = \sqrt{\frac{(1.6 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(9.11 \times 10^{-31})r}} \quad \text{M1}$$

$$= \frac{15.9}{\sqrt{r}} \quad \text{A0}$$

$$(ii) \quad v = \frac{nh}{2\pi mr} = \frac{15.9}{\sqrt{r}}$$

$$n = 1 \text{ so } \frac{h^2}{4\pi^2 m^2 r^2} = \frac{(15.9)^2}{r}$$

C1

$$r = \frac{h^2}{4\pi^2 m^2 (15.9)^2} = 0.053 \text{ nm}$$

A1

(iii) Energy = potential energy + kinetic energy

$$= -\frac{e^2}{4\pi\epsilon_0 r} + \frac{1}{2}mv^2$$

$$= -\frac{(1.6 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(5.3 \times 10^{-11})} + \frac{1}{2} (9.11 \times 10^{-31}) \frac{(15.9)^2}{(5.3 \times 10^{-11})}$$

C1

$$= -2.17 \times 10^{-18} \text{ J}$$

M1

$$= -\frac{2.17 \times 10^{-18}}{1.6 \times 10^{-19}} \text{ eV}$$

$$= -13.6 \text{ eV}$$

A0

(e) (i) For minimum wavelength, n is infinity

$$\lambda = \frac{1}{1.097 \times 10^7} = 91.2 \text{ nm}$$

A1

$$(ii) \quad E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{\lambda}$$

C1

$$E\lambda = 1.99 \times 10^{-25} \text{ J m}$$

$$= 1.99 \times 10^{-25} \text{ J (x}10^9\text{) nm}$$

M1

$$= 1.99 \times 10^{-16} \text{ J nm}$$

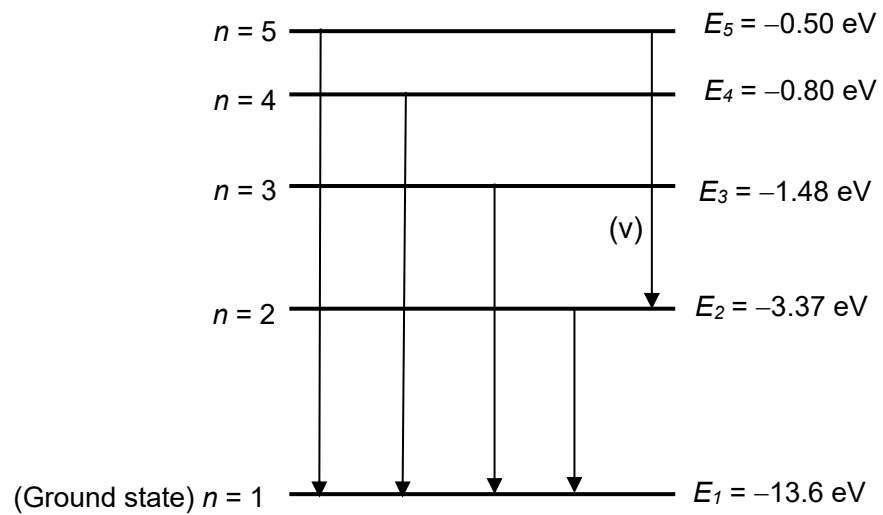
A0

(iii)

wavelength λ / nm	$E = \frac{hc}{\lambda}$ / eV
121.6	10.23
102.6	12.12
97.3	12.8
95.0	13.1

energy in eV A1

(iv)



5 energy levels shown, with progressively smaller energy difference between higher energy levels **C1**

Correct energy values for each level **C1**

Correct electron transitions from higher to lower energy levels **C1**

(v) For $\lambda = 434.1 \text{ nm}$, $E_h - E_l = 2.87 \text{ eV}$.

Hence the transition is from $n = 5$ to $n = 2$

A1